

Sophia N. Wilson

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Biography

Sophia N. Wilson is a Ph.D. fellow at the Sustainable Artificial Intelligence for Sciences (SAINTS) Lab, Department of Computer Science, University of Copenhagen (UCPH), supervised by Raghavendra Selvan and Erik Dam. Her research spans two strands: a technical strand developing algorithms and AI systems that are more resource efficient in data, compute, and energy use; and a socio-technical strand measuring and mitigating sustainability harms in the AI industry, including environmental impact and poor labour conditions. She is also active in bridging academia and industry, and in outreach promoting STEM fields, particularly to girls and young women.

Education

- 2025–present **PhD Fellow, Sustainable AI**, SAINTS Lab, Dept. of Computer Science, UCPH, Denmark.
Thesis: Glean: Towards Green Lean Data in the Era of Big Dark Data
Supervision: Raghavendra Selvan and Erik Dam.
- 2023–2025 **MSc in Computational Physics**, Niels Bohr Institute, UCPH, Denmark.
Thesis: "Quantifying the Reduction in Carbon Footprint of Physics-Informed Machine Learning".
Supervision: Raghavendra Selvan and Jens Hesselbjerg Christensen.
- 2019–2023 **BSc in Physics**, Niels Bohr Institute, UCPH, Denmark.
Thesis: "Measuring Cosmological Parameters with Fast Radio Bursts".
Supervision: Kasper Heintz and Johan Fynbo.

Selected Publications

In Conference Proceedings

- 2026 **Sophia N. Wilson**, Andrew Millard, Guðrún Fjóla Guðmundsdóttir, Raghavendra Selvan, and Sebastian Mair. Position: Stop Preaching and Start Practising Data Frugality for Responsible Development of AI. In *Proceedings of the 43rd International Conference on Machine Learning (ICML), Position Paper Track*, 2026.
- 2026 **Sophia N. Wilson**, Sebastian Mair, Mophat Okinyi, Erik B Dam, Janin Koch, and Raghavendra Selvan. How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI. In *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2026.
- 2026 **Sophia N. Wilson**, Jens Hesselbjerg Christensen, and Raghavendra Selvan. Performance-Carbon Trade-Offs across Architectural Biases in Shear Flow Forecasting. In *Proceedings of the 19th European Conference on Computer Vision (ECCV), LIMIT Workshop*, 2026.

Preprints

- 2026 Pedram Bakhtiarifard, **Sophia N. Wilson**, Mahmoud Afifi, Jonathan Wenshøj, and Raghavendra Selvan. Characterizing Learning in Deep Neural Networks using Tractable Algorithmic Complexity Analysis, 2026.

Research Projects

Dept. of Computer Science, UCPH

- Spring 2026 **Connecting Algorithmic Complexity and Deep Neural Networks.**
Collaborative work developing QuBD (Quantized Block Decomposition), a scalable method for estimating the algorithmic complexity of deep neural network weights, providing the first scalable characterisation of the learning-as-compression hypothesis for deep neural networks.
- Fall 2025 **Costs of Data Abundance and Benefits of Data Frugality.**
Led two studies: (1) quantified the environmental, social, and economic costs of hyper-datafication through large-scale metadata analysis of 550,000 Hugging Face datasets, a survey of data worker conditions in Kenya, and an examination of data centre investment and geographic concentration, published at FAccT 2026; (2) identified a preach-practice gap in the data frugal literature where energy savings are frequently cited as motivation but rarely measured, and developed recommendations for closing it, published at ICML 2026 (position paper track).
- Niels Bohr Institute & Dept. of Computer Science, UCPH
- Fall, 2024 – Spring, 2025 **Quantifying the Reduction in Carbon Footprint using Physics-Informed ML.**
Completed my MSc thesis applying physics-informed ML to reduce energy consumption in PDE simulations.

International Experience

- Aug. 2026 **Cetinje, Montenegro**, Eastern European Machine Learning Summer School (Best Poster).
- Feb. 2026 **Melbourne, Australia**, Machine Learning Summer School.
- Spring 2024 **University of Amsterdam, The Netherlands**, Exchange Semester.
- Aug. 2022 **University of Montréal, Canada**, Astromatic Machine Learning Summer School.
- Aug. 2022 **Nordic Optical Telescope, La Palma, Spain**, Summer School.

Teaching Experience

- 2026 **Data Analysis and Machine Learning**, MSc course for high school teachers, Dept. of Computer Science, Aarhus University.
- 2025 **Machine Learning A**, MSc course, Dept. of Computer Science, UCPH.
- 2025 **Mathematics A**, high school, Falkonergårdens Gymnasium.
- 2023 **Summer Science Camp**, elementary school, Insero.
- 2022-23 **Mathematics for Physicists 2**, BSc course, Niels Bohr Institute, UCPH.
- 2022-23 **Mathematics for Physicists 1**, BSc course, Niels Bohr Institute, UCPH.
- 2020-21 **Mechanics and Special Relativity**, BSc course, Niels Bohr Institute, UCPH.

Supervision

- 2026 **Co-supervised two BSc students (with R. Selvan)**, *"The Environmental Impact of Generative AI: A Life Cycle Assessment as a Basis for Studying Consumer Perception"*, Department of Communication, UCPH.

Talks

- Sep. 2026 **Center for AI, Danish School of Media and Journalism**, Webinar, "Bæredygtig AI: Udfordringer og Løsninger".
- May 2026 **Pioneer Centre for AI, UCPH**, Workshop on Green AI, presenting "How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI".
- May 2026 **National Center for AI in Society, UCPH**, Workshop on Sustainable AI, "On the Challenges in Assessing the Sustainability of AI" (joint with R. Selvan).

- May 2026 **Technical University of Denmark, Visual Computing Group**, Presenting "How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI".
- Apr. 2026 **Technical University of Denmark, Embedded AI Group**, Presenting "Stop Preaching and Start Practising Data Frugality for Responsible Development of AI".
- Apr. 2026 **Lund Stem Cell Center, Lund University**, Workshop on Sustainable AI, "How to measure and manage energy consumption of AI models".
- Aug. 2025 **D3A**, Workshop on Resource-Aware Machine Learning, "Quantifying the Reduction in Carbon Footprint of Physics-Informed Machine Learning".
- Apr. 2025 **Copenhagen Center for Social Data Science, UCPH**, "On the Carbon Footprint of AI: From Monitoring to Mitigation" (joint with R. Selvan).
- Mar. 2025 **Niels Bohr Institute, UCPH**, 4th Annual NBI MSc Student Symposium, "Quantifying the Reduction in Carbon Footprint of Physics-Informed Machine Learning".

Panels

- Apr. 2026 **Lund Stem Cell Center, Lund University**, "Sustainable AI in Practice: From Computational Costs to Real-World Impacts".
- Jun. 2026 **Folkemødet, Bornholm**, "Kærlighed, Klimakrise og Kunstig Intelligens".

Guest Lecturing

- Feb. 2026 **University College Copenhagen**, BSc course on Ethics and Technology, "AI, Resource Use, and Sustainability".
- May 2025 **Department of Computer Science, UCPH**, PhD course on Machine Learning for Sciences, "Physics-Informed Machine Learning as a Carbon-Efficient Approach".

Outreach

- Sep. 2025 **Det Åbne Gymnasium**, "Energiforbrug og CO₂-aftryk af kunstig intelligens".
- May 2025 **Falkonergårdens Gymnasium**, "Hvordan er det at læse fysik på universitetet?".
- 2021-24 **Women in Physics**, Contributing to initiatives promoting gender balance in STEM.

Organisation

- Mar. 2026 **SAINTS Lab, Department of Computer Science, UCPH**, Co-organiser and moderator, SAINTS'26 Workshop on Sustainable AI for Sciences.

Media

- Dec. 2025 **Interview by Women in Physics.**
<https://kvinderifysik.dk/2025/12/15/interview-with-sophia-wilson/>